LENNARD JOHN P. ROBENTA

COMPUTER ENGINEERING STUDENT

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Computer engineering student at Mapua University, with strong experience in embedded systems, machine learning, and computer vision. Developed an AI-based medical imaging and classification system using Python, IoT projects using C++, Arduino, and ESP microcontrollers, cloud integration services such as Google Firebase and ThingSpeak. I'm passionate on building systems that combine AI and hardware technologies.

TECHNICAL SKILLS

- Programming: Python | C++
- Machine Learning: Image Segmentation (U-Net Architecture) | SVM | Feature Extraction
- Embedded Systems: Microcontroller | Sensor integration and real-time data acquisition| Actuator control
- IoT & Cloud Integration: Google Firebase | Data visualization using ThingSpeak

AREA OF INTEREST

Prototyping | Embedded Systems | IoT | Database | Research | Machine Learning | Cloud Integration

PROJECTS

AI-Based Periodontitis Severity Classification (Thesis) **Oct 2024 - Present**

Technologies: Python, Machine Learning, Image Segmentation

- Developed an automated system to classify the severity of periodontitis using dental radiographs.
- Processed and analyzed 3000+ dental radiograph images from Kaggle datasets.
- Designed algorithms to segment teeth and their parts, measure radiographic bone loss, and periodontal indicators from segmented images.

RFID and QR Library Borrowing System **July 2024 - Present**

Technologies: Arduino, RFID, QR Scanner, IoT

- Developed a library management system using RFID for users and QR codes for book information Sent borrowing
- records to cloud storage using Google Firebase via ESP-01S.
- Developed a user interface using an OLED and buttons.

SanSenSys: Sanitizing by Sensing System **November 2025**

Technologies: Arduino, Sensors, Actuators, ESP8266, IoT

- Developed an infrared-based contactless hygiene system that automatically dispenses hygiene liquids
- Implemented liquid level monitoring using ultrasonic sensors with LED warning indicators
- Integrated real-time environmental monitoring using DHT22 and an RTC module, displayed on an LCD.

HRMS – Hotel Reception Management System **July 2025**

Technologies: Arduino, Sensors, Actuators, Seven-Segment Display (SSD)

- Developed a hotel prototype integrating smart lobby appliances management, motor-powered password-protected vault security, parking management, and traffic control.
- Implemented smart parking and toll gate automation using infrared sensors and a four-digit SSD.
- Integrated a fire detection system that triggers alarms and suspends non-critical operations.

Project AGOS – Add-on Garden Automation System **March 2025**

Technologies: Arduino, Sensors, Actuators, Seven-Segment Display (SSD)

- Designed an automated irrigation system capable of monitoring five plant pots independently
- Implemented a soil moisture-based servo-controlled irrigation system
- Developed a multi-display monitoring interface for environmental and water level data using an LCD and four-digit SSD with LED and buzzer for alarm

EDUCATION

Bachelor of Science in Computer Engineering **August 2021 - Present**

Mapua University

- Specialization: Embedded Systems
- Thesis: "Severity Classification of Periodontitis Using SVM with A-Res-UNet Image Segmentation"
- Design: "RFID and QR Code-Based Library Book Borrowing Management System Using ATmega328P"

COURSERA CERTIFICATIONS

Introduction to DevOps | **Deep Learning with PyTorch : Image Segmentation** | **Introduction to Networking and Storage** | **Introduction to Cybersecurity Essentials** | **Introduction to Software, Programming, and Databases**